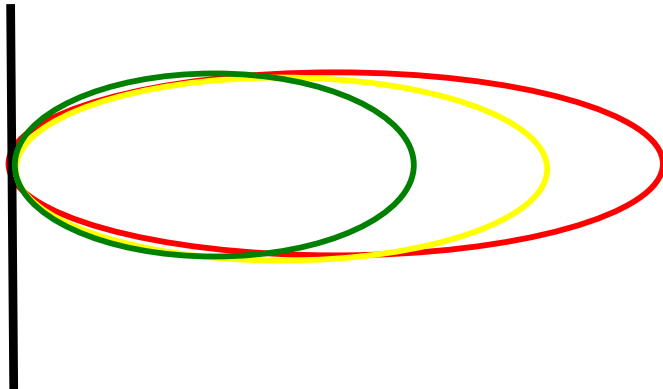


L6

Passive Tension

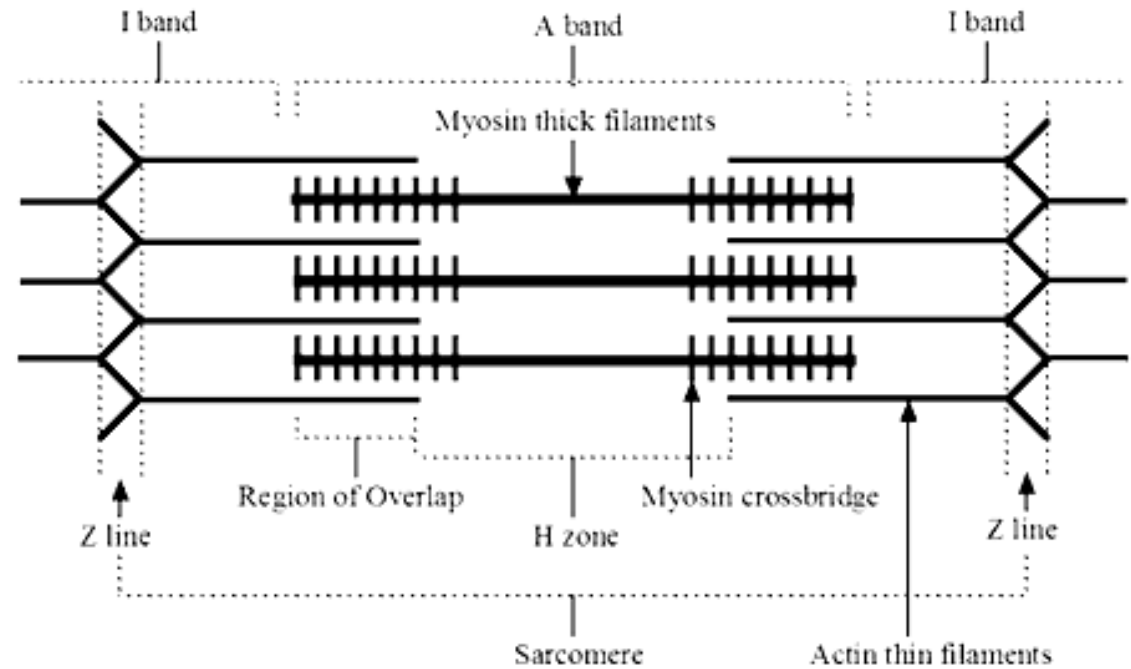


<https://depositphotos.com/vector-images/slingshot-cartoon.html>



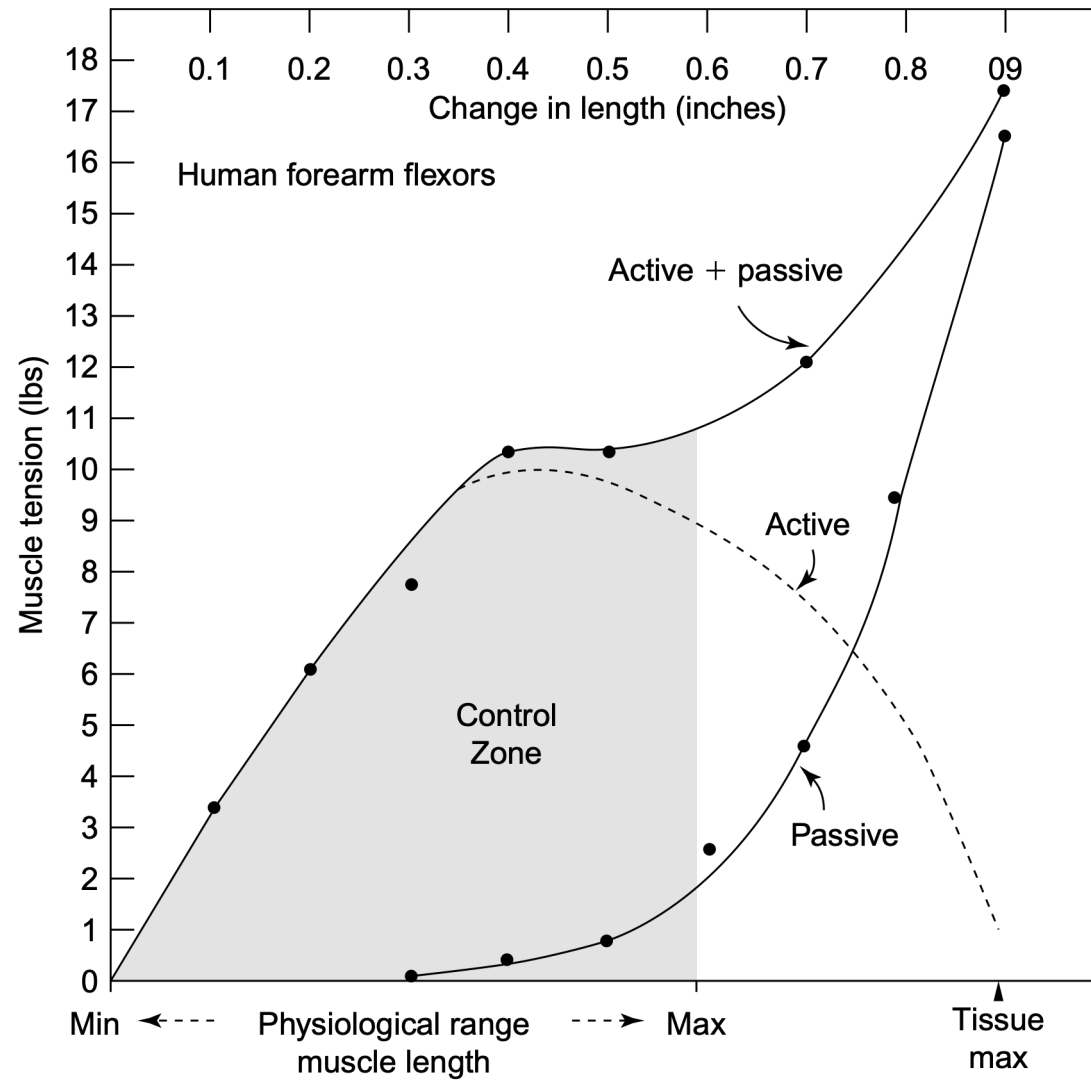
No Energy

Active Tension

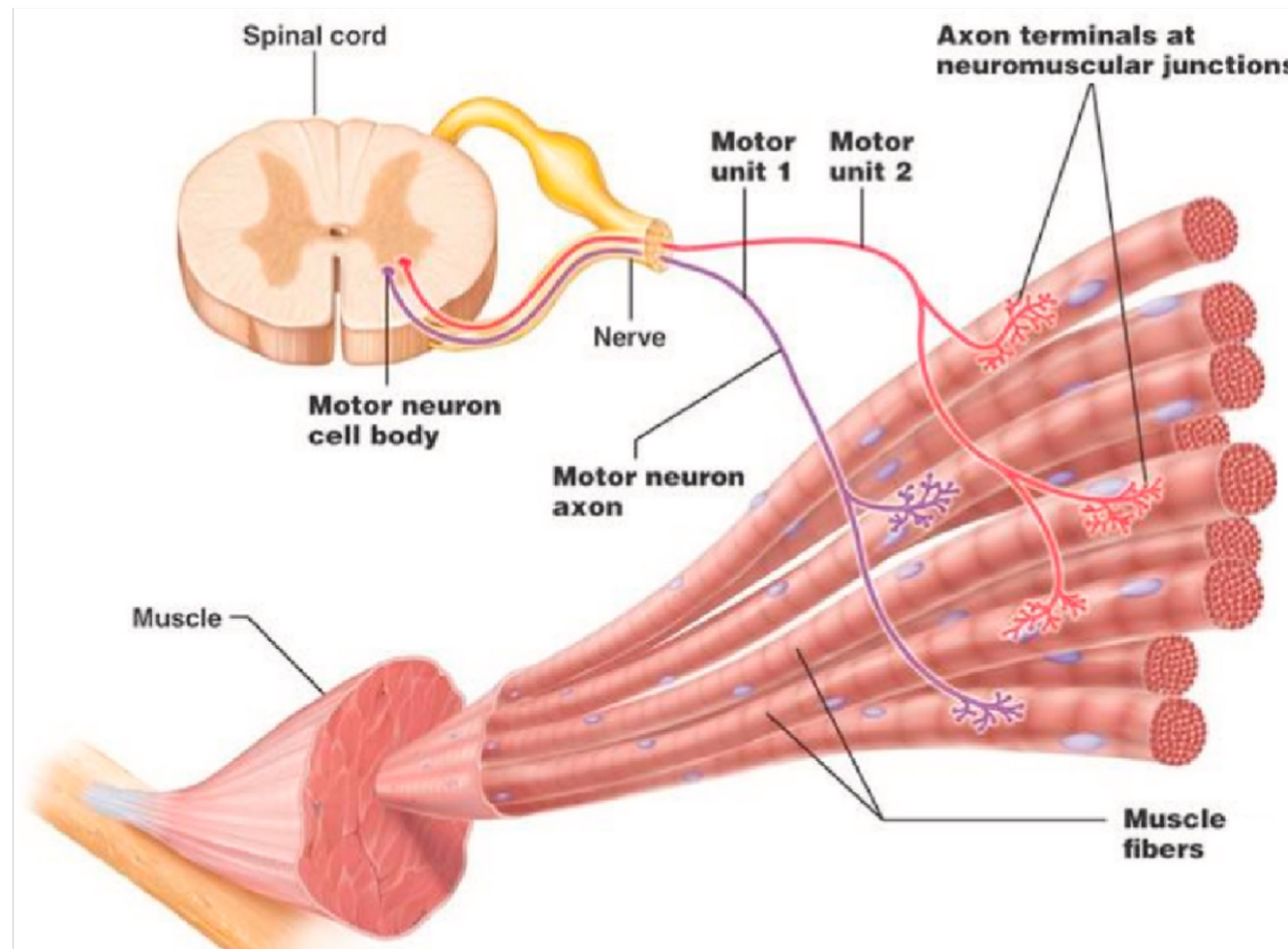


Energy required

Total tension



Motor Unit

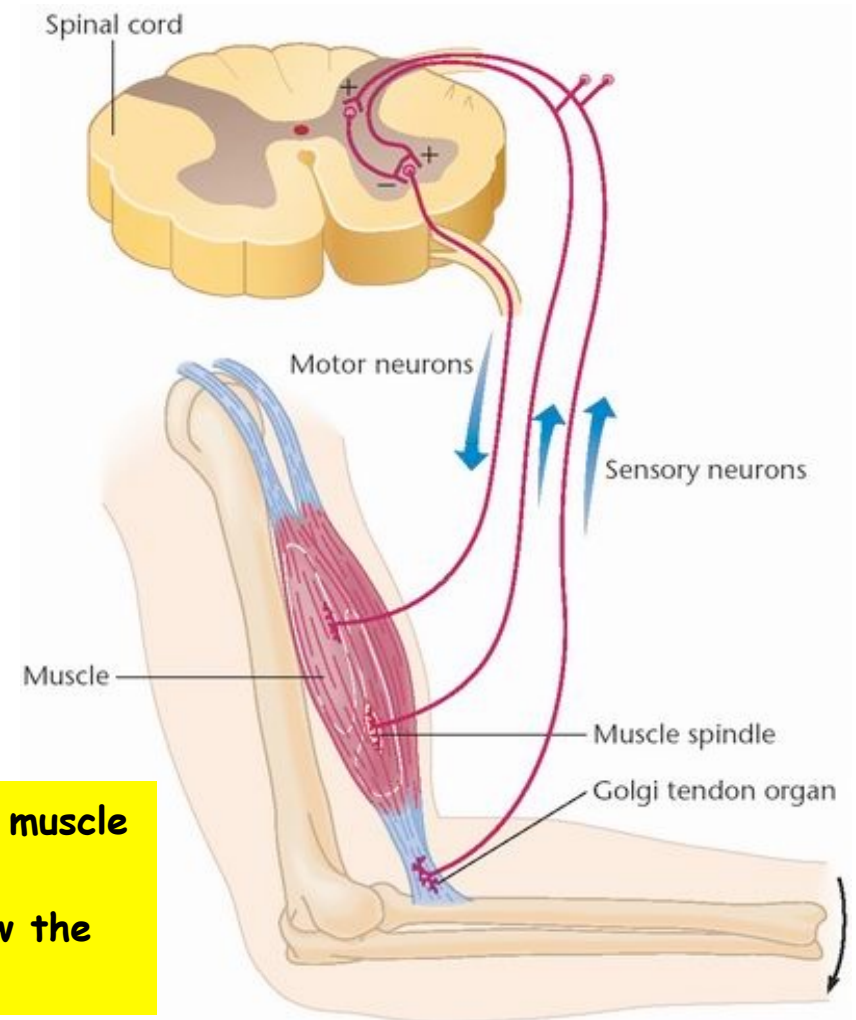


Proprioception

Perception of information from receptors present in the muscles, joints, tendons and skin.

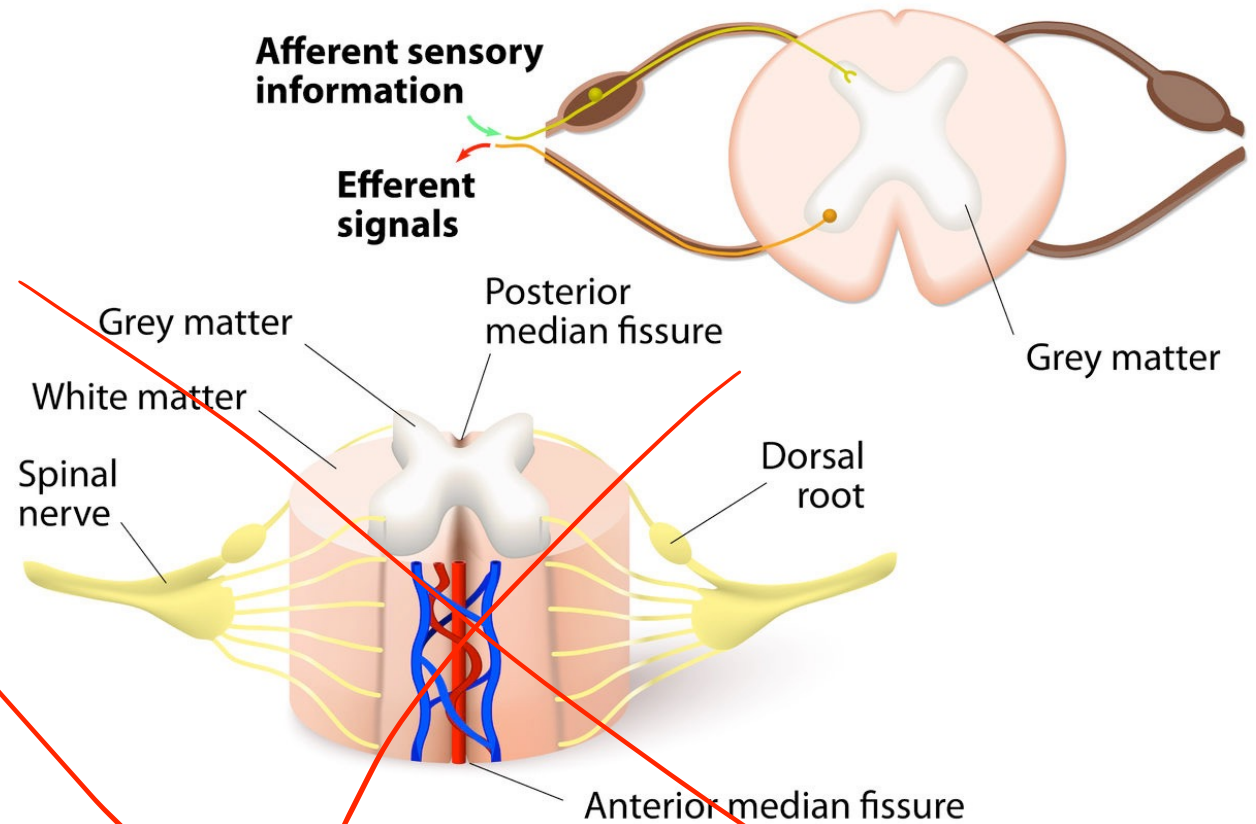
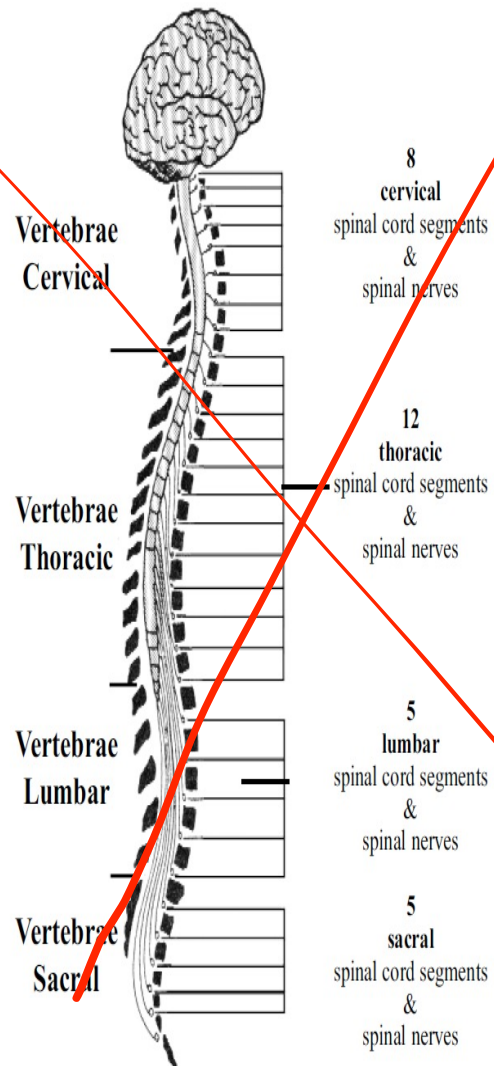
Kinesthesia

- Muscle spindles
- Golgi tendon organs
- Joint receptors
- Cutaneous receptors



**** In the book there is lot of information related to the muscle types contributing to these proprioreceptors. You do not have to remember that. Its important to know the basic functions of these proprioreceptors.**

Spinal Cord

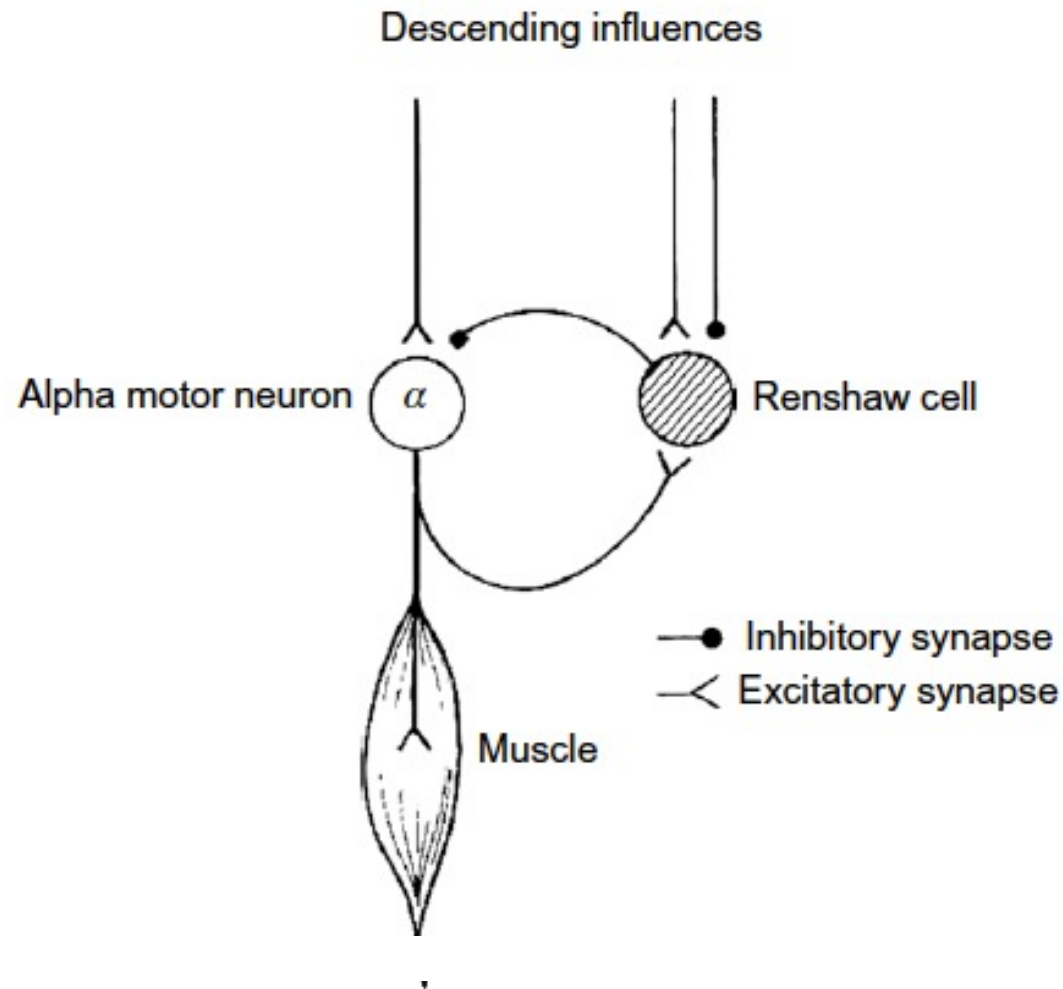


<https://www.spineuniverse.com/anatomy/spinal-cord-nerves-brain>

Spinal Cord

- Relay center
 - Recurrent connections
 - Reciprocal connections
- Reflex action

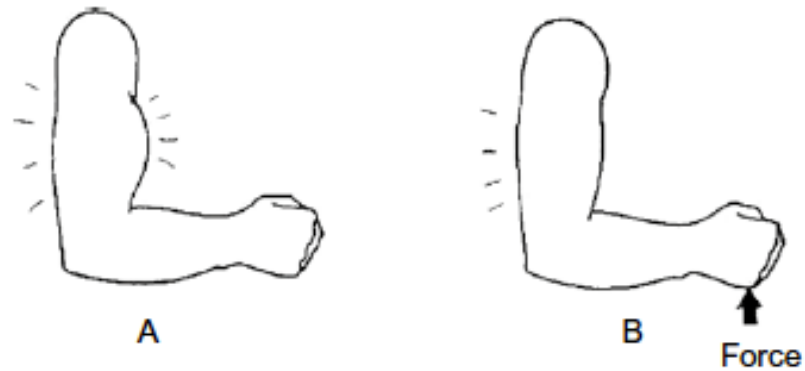
Recurrent connections



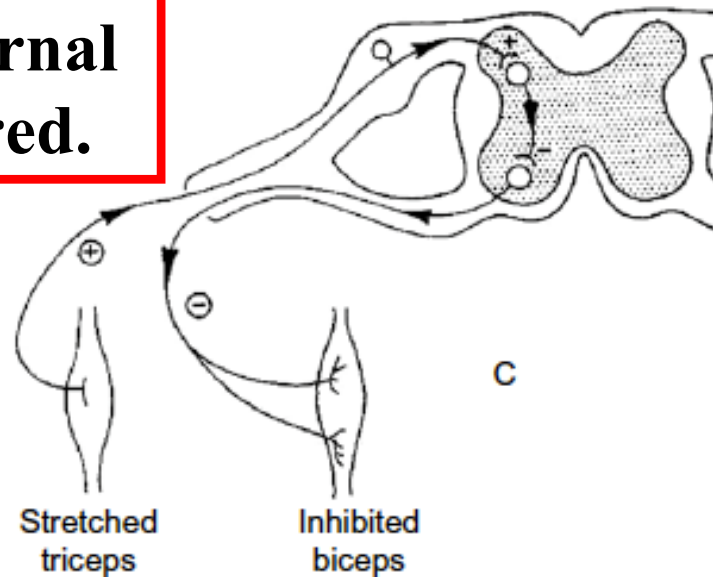
Affects sensitivity of motor neurons

Reciprocal connections

Co-contraction



Activated when external loads are encountered.



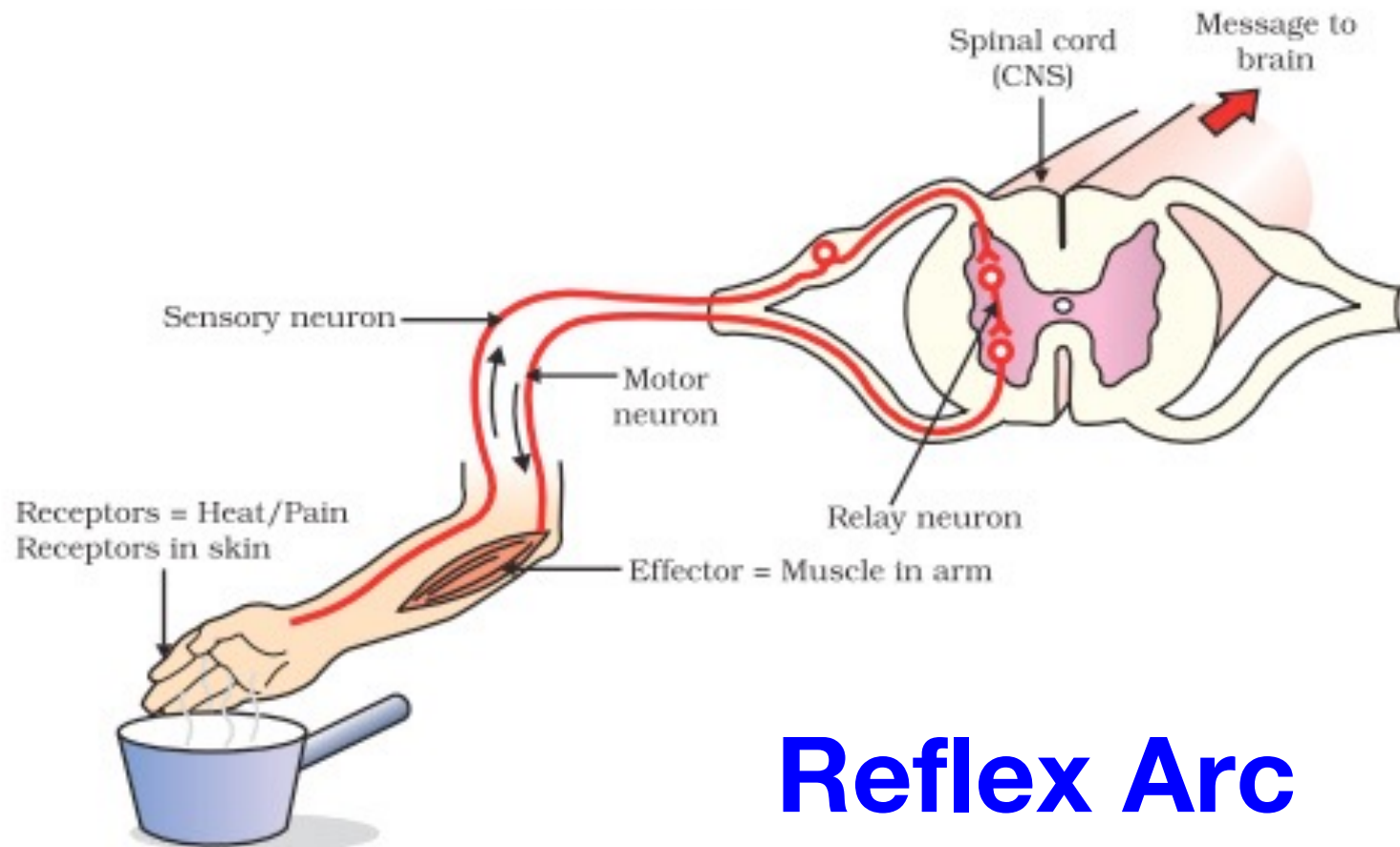
Reciprocal Inhibition

Reciprocal Inhibition

- **Co-activation of muscles is possible.**
- **Activated when external loads are encountered.**
- **Muscles do not directly inhibit each other.**

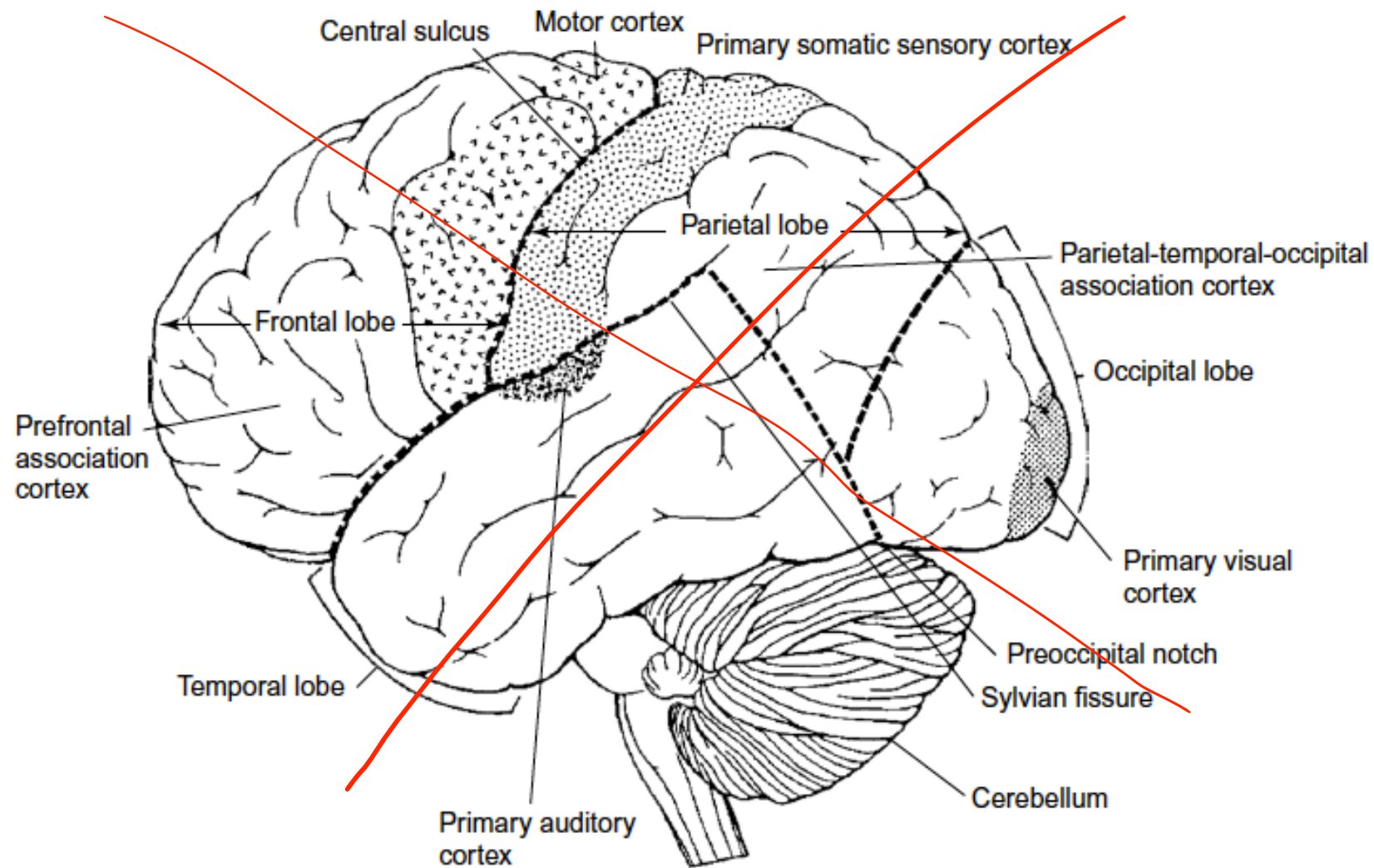
Why?

Reflex Action

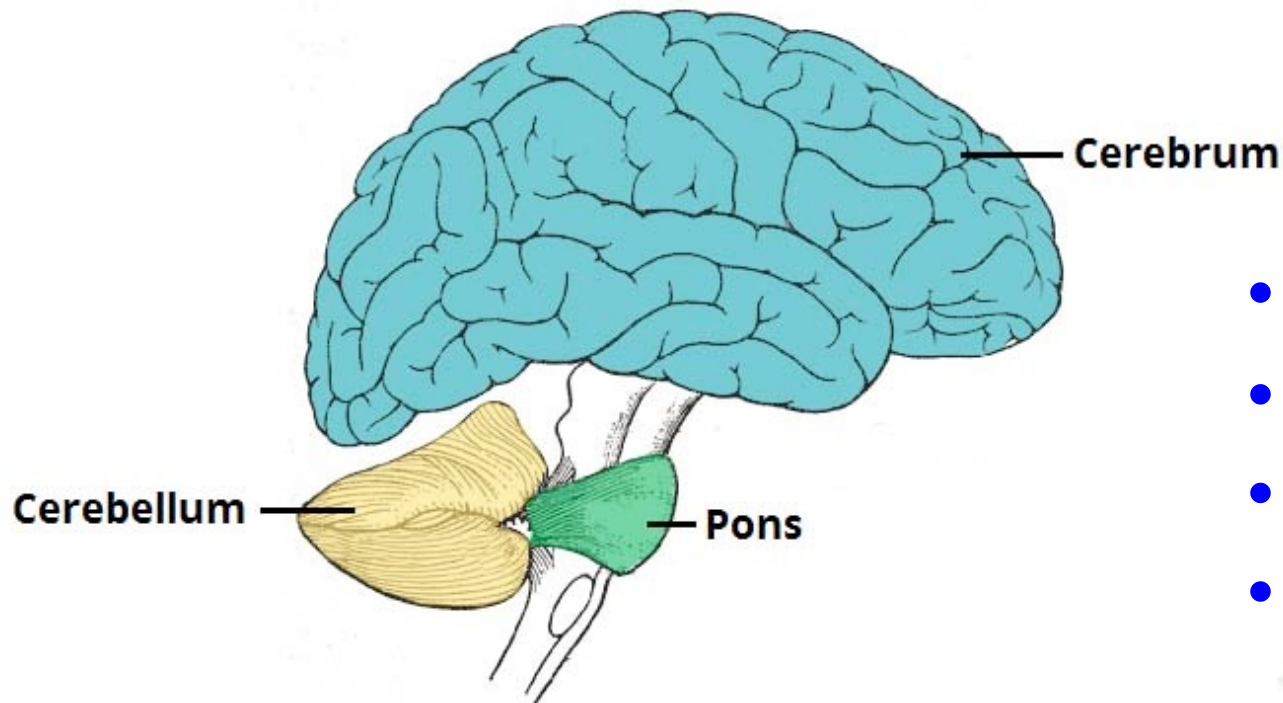


Reflex Arc

Brain Regions in Motor Control



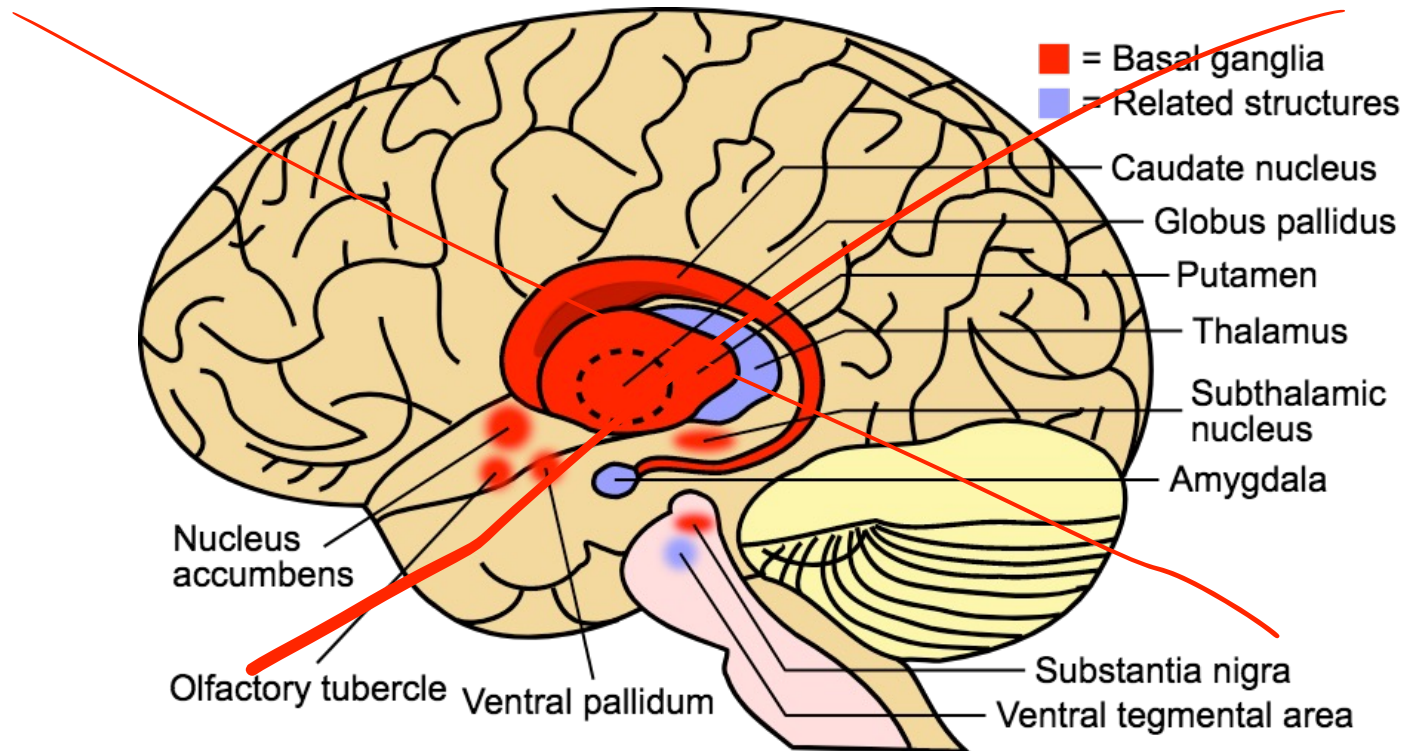
Cerebellum



- **Muscle Tone**
- **Coordination**
- **Timing**
- **Learning**

<https://teachmeanatomy.info/neuroanatomy/structures/cerebellum/>

Basal ganglia



https://en.wikipedia.org/wiki/Basal_ganglia

- Retrieval of movement plans
- Scaling amplitudes of movement
- Perceptual motor integration

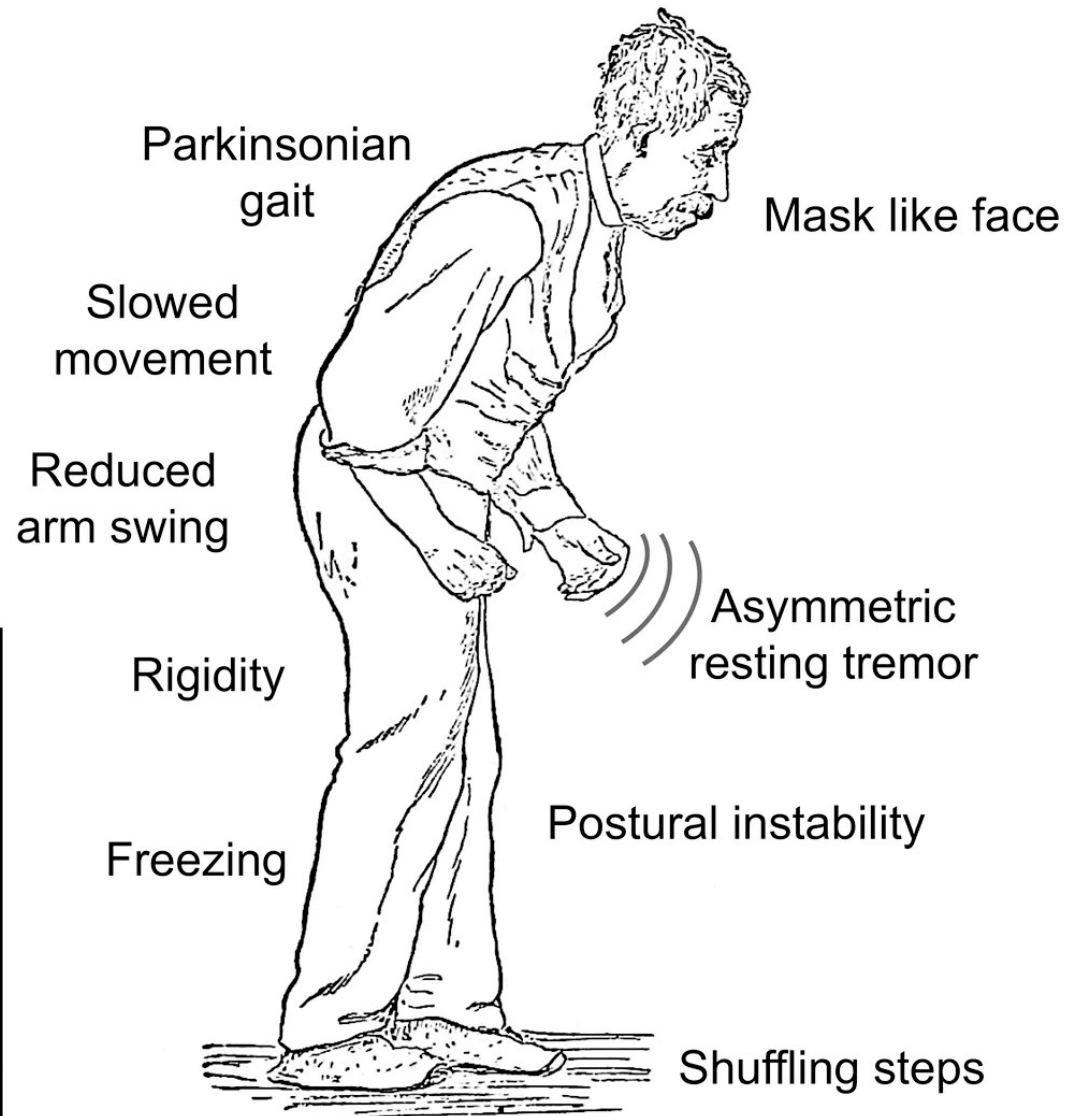
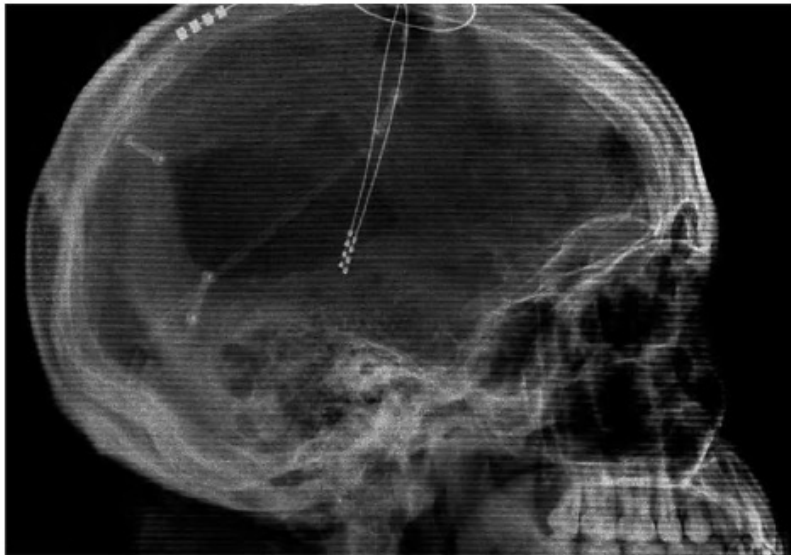
Huntington's Disease

- **Moodiness**
- **Fidgeting**
- **Depression**
- **Muscle twitching**
- **Personality shifts**
- **Cognitive decline**
- **Dementia**
- **Hallucinations**
- **Involuntary movement (Chorea)**
- **Speech difficulties**

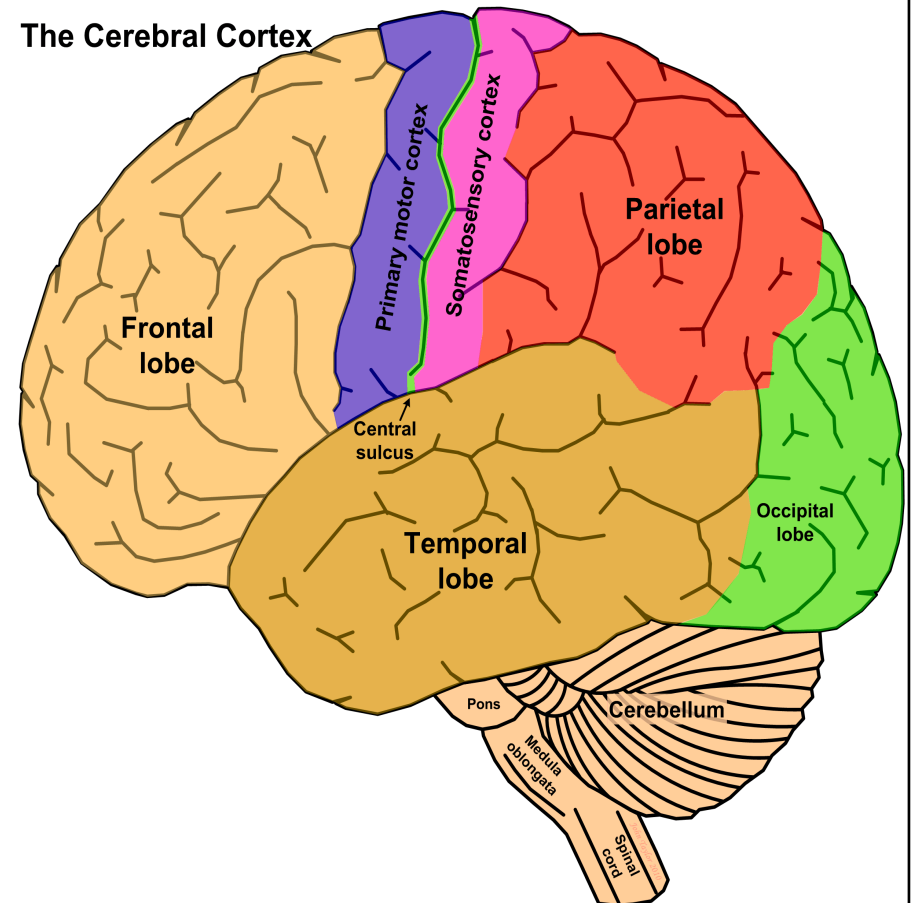
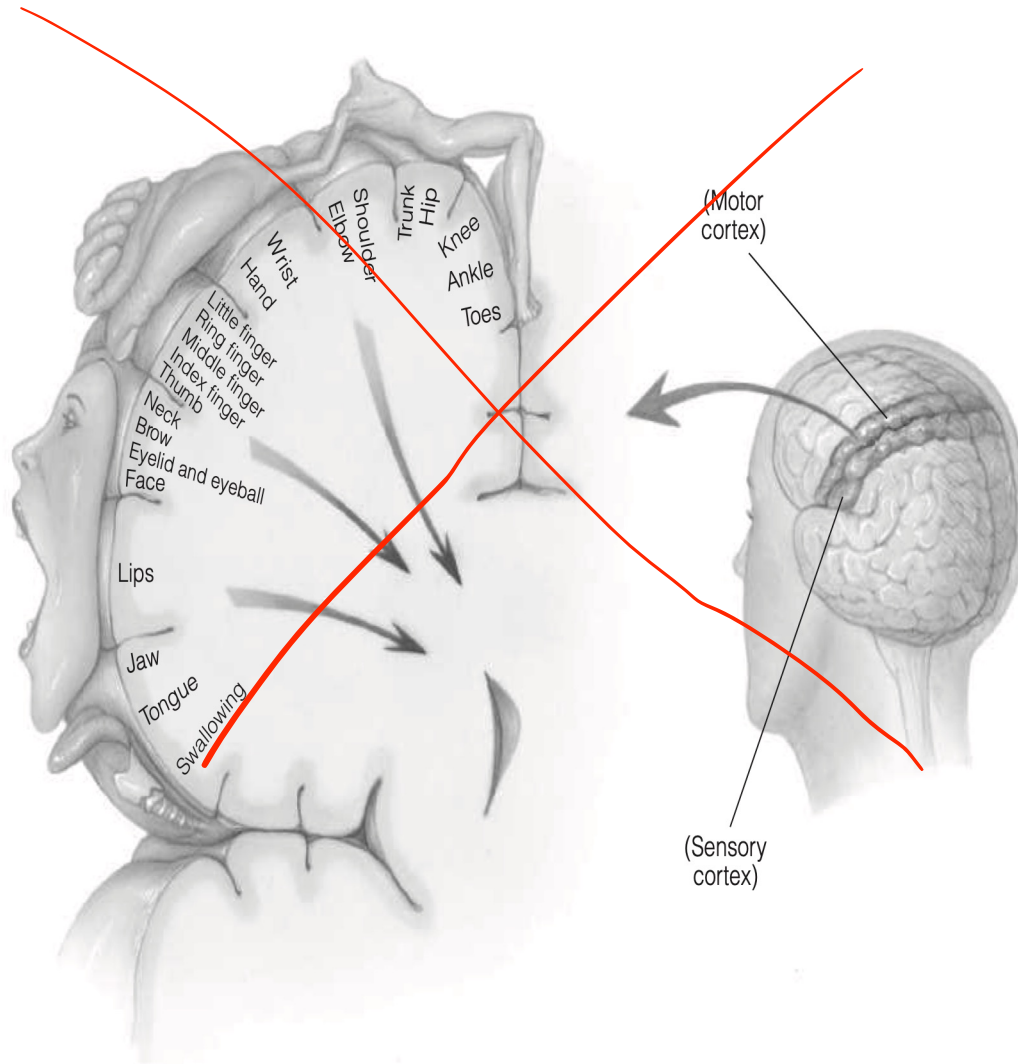
Parkinson's Disease

- Resting tremors
- Akinesia
- Bradykinesia
- Rigidity

Deep Brain Stimulation



Primary Motor Cortex



Homunculus



<https://www.amazon.co.uk/The-Original-Homunculus-Company-Sensory/dp/B076YBYHMN>